BAYMAX Experiment Notebook

This folder explains the experiments, rather than merely listing result filenames. Each numbered note covers one training experiment, inference experiment or clearly identified measurement/audit investigation. Multiple evaluations of the same checkpoint can live in its note; a 30-case diagnostic does not become a separate trained model.

Start with a question or follow the stages below. Numbers within folders are reading order, not invented run IDs or a guarantee of exact historical chronology. Every evaluation table links to its original JSON and records its saved date.

Where We Stand

The selected development reference is cleaned anchored SFT at two epochs. It is the reference for balanced routing, not the winner of every metric.

Candidate	Task successes / 150	Routing macro-F1	Invalid outputs / 150	Premature ACT on boundary cases
Anchored two-epoch SFT	44	0.514	2	31
DPO from that reference	53	0.376	5	56
2:1 SFT	29	0.477	8	22
DPO from 2:1 SFT	31	0.453	7	34
DPO with completed ACT rejections	46	0.482	7	45
Rank-16 two-epoch SFT	46	0.450	7	42
Lower-learning-rate two-epoch SFT	45	0.436	15	37
QKVO two-epoch SFT	52	0.437	14	40

Those are measured development results, not evidence of a robust overall win. First DPO has the highest task completion but substantially worse abstention. Completing its missing rejections recovers some balance, yet remains below original SFT macro-F1. Rank 16 also worsened routing and validity despite two additional task successes. Lower learning rate did not improve balance either. QKVO completes more tasks but has worse routing and output validity: retain it as a task-completion candidate, not the balanced reference. A's original rank-8 recipe remains the reference. No model switch or new training method was implemented by this documentation task.

The full recipes, matrices, sources and caveats are in the individual notes. For the most recent sequence, read the two-epoch reference, four-epoch trial, matched contrasts, 2:1 trial, and the three DPO notes in that order.

How an Experiment Flows

1. Author or import training examples under the general policy and tool contracts.
2. Protect source train/validation/test membership and contrast/content groups.
3. Mix and sample a deterministic raw training cap.
4. Render the chat template, tokenize, mask non-assistant labels and truncate.
5. Inspect the usable artifact that actually reaches the trainer.
6. Train a new versioned adapter.
7. Run the chosen adapter on a separate evaluation suite through fake adapters.
8. Read task outcome, routing, output validity and wording metrics separately.

Pipeline fixes at steps 2-5 or 7-8 can change the validity of the evidence without creating a better trained model. That distinction explains several apparent jumps in the history.

Vocabulary

Term	Meaning in these notes
ACT	Select supported productive tools when the request is safe and complete.
CLARIFY	Supported safe action needs missing/ambiguous information or required confirmation.
REFUSE	Action/capability is unsafe, unavailable, unsupported or out of scope.
Meta-tool	request_clarification(question) or refuse_request(reason); not an external mutation.
xLAM	External Salesforce function-calling dataset hosted on Hugging Face.
BAYMAX-control	Authored training data; "control" is the dataset name, not necessarily a baseline model.
Mixed training	xLAM plus authored BAYMAX examples.
SFT	Learn supplied correct assistant answers using target-token cross-entropy.
LoRA	Train small adapter weights rather than updating every base-model weight.
DPO	Learn to prefer a chosen answer over a rejected answer for the same prompt.
Anchored sampling	Keep reference examples/positions and replace only explicitly eligible rows.
Matched contrast	Closely related requests/states/tools with a genuine decision-changing difference.
Raw cap	Number selected before tokenization and assistant-label filtering.

Term	Meaning in these notes
Usable rows	Examples retaining trainable assistant labels after preprocessing.
Epoch	One pass over the usable training artifact; not one optimizer step.
Exact duplicate	Identical example content; not a test of semantic or template similarity.
Reference freeze	Record hashes to detect drift; it does not lock or copy files.

Metrics Are Different Questions

- **Task success:** did all scenario checks pass, including arguments and message criteria?
- **Routing macro-F1:** average ACT, CLARIFY and REFUSE F1 equally. F1 combines precision (correctness of predictions of that class) and recall (coverage of expected cases of that class). Invalid/unknown outputs miss their expected class.
- **Decision accuracy:** correctness among recorded known decisions. Read it with coverage; unknown outputs are not included in its conditional denominator.
- **Clarification/refusal accuracy:** required question/reason keyword matches. A correct label can still have an incomplete or misleading explanation.
- **Invalid-output rate:** strict parser rejections, separate from valid but inappropriate decisions or tool-execution failures.
- **Hallucination rate:** currently unavailable tool names, not every invented person, date or success claim. Invalid parsing can leave no accepted calls to count.
- **Latency/cost:** local reported API cost zero is not zero electricity, hardware or training cost. Replayed results inherit original inference timing.

Tables use ratios from zero to one, rounded to three decimals, unless they show explicit counts. `n/a` means unavailable or inapplicable, not zero.

Which Evaluation Suite?

Suite	Size	Purpose
v1	50	Original benchmark, used by early trials.
Combined v1 + v2	150	Established development comparison. Current decision support is ACT 81, CLARIFY 40, REFUSE 29.
Template shift	30	Separate wording/structure diagnostic: ten cases per decision.
Refusal subset	7	Small single-class diagnostic, not balanced overall evaluation.
Smoke subsets	1-4	Sanity checks; not estimates of full benchmark quality.

Historical policy-era fixtures/scoring differ from current measurements. Do not combine the template-shift and established-suite rates into one score. Their repeatedly inspected status means neither is an untouched final benchmark.

Early Training

The first external-only, synthetic-only and mixed trials. Historical provenance is incomplete.

- External Tool Data Only: First Local Fine-Tune
- BAYMAX Control Data Only
- Control Data With Contrast Pairs
- Control Data With Targeted Refusals
- Original External and BAYMAX Training Mix
- Expanded Control Generator: Early Mixed Trial
- Weighted Control Examples: Early Oversampling Trial
- Clarification and Refusal as Explicit Meta-Tools

Format, Prompt and Sampling

Input alignment, weighting, unique-first caps and inference-only prompt changes.

- Aligning Training and Evaluation Formats
- Extra Clarification and Refusal Repeats
- Legacy Boundary-Weighting Trial
- Fixed Unique Boundary Pool at 1,000 Rows
- Increasing the Cap to 1,500 Without Repeating Boundaries
- First Policy Prompt: Evaluation Without Retraining
- Expanded Policy Prompt: Evaluation Without Retraining
- Returning to the Recorded Training Prompt

Data Quality and Class Mix

Changes in class quantities, partition integrity, policy, credentials and target compatibility.

- One Hundred ACT, CLARIFY and REFUSE Examples
- Early Balanced BAYMAX Data at Two Epochs
- Schema-Diverse Control Examples
- Increasing Productive BAYMAX Examples to Two Hundred
- Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples
- Content-Group Splits With Seventy Percent ACT
- Corrected Policy, Credential Filtering and Representative Validation
- Corrected Policy With Two Hundred Productive Examples

- Target-Contract Cleaning With Resampled External Rows
- Target-Contract Cleaning With Anchored External Rows

Training Duration and Matched Contrasts

Fresh one/two/four-epoch comparisons, matched groups and the 2:1 SFT ratio.

- Anchored Data at Two Epochs: Selected Reference
- Anchored Data at Four Epochs
- Matched Contrasts at Fixed Class Counts
- Two-to-One Productive-to-Boundary SFT Ratio

SFT Hyperparameters

Controlled adapter/training changes from the selected SFT reference. B, C and E are completed. Increasing rank, lowering the learning rate and adding attention projections did not beat A's balanced routing. E improved task completion, with output-contract failures documented separately.

- Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe
- Experiment C: Lower Learning Rate at Two Epochs
- Experiment E: LoRA on All Four Attention Projections

Preference Training

The original DPO stage, alternative SFT initialization, and completed ACT rejections.

- DPO From the Selected Two-Epoch SFT Reference
- DPO From the Two-to-One SFT Model
- Completing Missing Productive Rejections for Boundary Decisions

Measurement and Data Audits

Baseline interpretation and engineering corrections. These are not new model-training wins.

- Untrained Qwen: What the Baseline Actually Measures
- Effective Token Distribution and Training-Cap Audit
- Strict Parser Replay: Correcting Measurement, Not Training
- Training Data Contract and Policy Review
- Decision Metrics Separate From Explanation Wording
- Clarification Output Contract: Target Audit and Failure Diagnostics

Rules for Reading Comparisons

1. Compare the same scenario suite, prompt, generation settings, parser and scorer.
2. Distinguish a new trained checkpoint from an inference-only prompt test or strict replay of saved generations.
3. Check actual usable rows and tokens. A 1,000-row raw cap can deliver 974 rows.
4. A 200/100/100 BAYMAX quota plus 600 xLAM ACT examples is 800/100/100 overall, not a balanced three-class mixture. Multi-tool sequences still count as ACT.
5. Matching broad quantities does not guarantee matching domain/subtype coverage.
6. A fresh four-epoch run is not resuming a saved two-epoch adapter. A changed cosine horizon also changes per-step learning rates.
7. A 705-pair and an 800-pair DPO run at one epoch have different training exposure.
8. Single-seed changes are observations, not statistical proof or established causes.
9. Do not turn an inspected evaluation failure into a fixture-specific training target.
10. Do not assign today's config defaults to an old checkpoint with missing provenance.

Reproducibility and Sources

The external source revision used by the pinned pipeline is `26d14ebfe18b1f7b524bd39b404b50af5dc97866`. The later matched training prompt is identified by `a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468`.

Configs can declare context 2,048/batch two/accumulation eight while owner low-VRAM commands actually use 1,024/one/16. Later notes state actual reported overrides. Earlier notes explicitly retain missing information instead of filling it by guess.

Original evidence remains in the result folder. That older catalog predates some recent runs; use these notes for their explanations. FILE_RENAMES.json resolves historical names without changing result bytes. Config links identify recipes; generated model/data links are local-only and may not exist on another checkout.

Methodology remains defined by BENCHMARKING.md and SCORING.md. Detailed investigations remain in TRAINING_DATA_AUDIT.md and DPO_EXPERIMENT.md. Some older status paragraphs describe preparations that later completed; measured result artifacts take priority.

This notebook covers all 65 saved Phase 2 evaluation JSON files present on 2026-10-10, plus early external-only local results and supporting audits. No raw result, training example, evaluation target or model weight was changed. No long training/evaluation/audit command was run. These notes are not staged, committed or pushed; the owner's restriction on uploading Phase 2 documents remains.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

External Tool Data Only: First Local Fine-Tune

Experiment index

Question and Context

Could ordinary function-calling SFT make local Qwen better at BAYMAX tasks before adding domain-specific clarification and refusal training?

This is the xLAM-only starting point. Its files live under Phase 1 because that is where the first local comparison was stored; it belongs in the training history nonetheless.

Setup

- Base model: Qwen/Qwen2.5-1.5B-Instruct.
- External source: Salesforce/xLAM-function-calling-60k, downloaded through Hugging Face.
- Owner's medium-run logs describe a 1,000-row training cap, 200 validation rows, one epoch, context 1,024, batch one and gradient accumulation 16.
- LoRA requested: rank 8, alpha 16, q_proj/v_proj, dropout 0.05, no bias.
- The original medium run reported NaN validation loss. A later fixed run was evaluated; the saved result names do not establish which precise changes explain the fix.
- Bare Qwen is the comparison, not the scripted baseline or hosted GPT model.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen: full suite	50	11/50 (0.220)	0.540	0.869	0.000	0.071
xLAM-only LoRA: full suite	50	12/50 (0.240)	0.480	0.854	0.028	0.000
xLAM-only LoRA: legacy full suite	50	12/50 (0.240)	0.480	0.854	0.028	0.000
Bare Qwen: one-case check	1	0/1 (0.000)	1.000	0.500	n/a	n/a
Bare Qwen: four-case smoke	4	0/4 (0.000)	0.250	0.875	0.000	n/a
xLAM-only LoRA: one-case check	1	0/1 (0.000)	1.000	0.750	n/a	n/a
xLAM-only LoRA: three-case smoke	3	0/3 (0.000)	0.000	1.000	0.000	n/a

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen: full suite: 2026-08-11 UTC, 50 cases.
- xLAM-only LoRA: full suite: 2026-08-11 UTC, 50 cases.
- xLAM-only LoRA: legacy full suite: 2026-08-11 UTC, 50 cases.
- Bare Qwen: one-case check: 2026-08-11 UTC, 1 cases.
- Bare Qwen: four-case smoke: 2026-08-11 UTC, 4 cases.
- xLAM-only LoRA: one-case check: 2026-08-11 UTC, 1 cases.
- xLAM-only LoRA: three-case smoke: 2026-08-11 UTC, 3 cases.

What We Learned

The full-suite gain was one additional task: 12/50 versus 11/50 for bare Qwen. Tool-call accuracy fell, and clarification/refusal remained very weak. This was evidence that training and local inference could run, not a convincing policy-learning result.

The one-, three- and four-case files are smoke diagnostics. They should not be compared to the full-suite rates as if they were additional full experiments.

Limits

These are historical parser/prompt measurements. Exact base revision and complete runtime provenance are not recoverable from these result files alone. The legacy and later full artifacts have the same aggregate success but different latency; that does not prove identical generations.

The NaN loss was a training diagnostic, not evidence that all later checkpoints were invalid. Do not infer a causal fix from filenames alone.

Supporting Files

- qwen2_5_1_5b_lora_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

BAYMAX Control Data Only

Experiment index

Question and Context

Would training directly on BAYMAX's tools and boundary decisions help more than generic external tool data?

"Control" here means the authored BAYMAX-control dataset. It does not mean an untrained experimental control.

Setup

Synthetic training requests covered productive actions, clarification and refusal, using BAYMAX interfaces rather than evaluation fixtures. This checkpoint was evaluated without mixing xLAM into its training source.

The current control config is a navigation aid, not proof of the exact settings used by this historical checkpoint. The saved evaluation files do not record a complete frozen training recipe.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Control-only	50	2/50 (0.040)	0.500	0.578	0.000	0.071
Control-only / early parser fix	50	10/50 (0.200)	0.520	0.806	0.083	0.071

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Control-only: 2026-08-22 UTC, 50 cases.
- Control-only / early parser fix: 2026-08-22 UTC, 50 cases.

Interpretation

The original evaluation recorded 2/50 successes. The early-parser-fix artifact recorded 10/50. That large difference makes measurement behavior part of the explanation: it cannot be credited wholly to learning.

Even under the later measurement, tool selection and boundary behavior were weak. Purely authored data did not establish a robust replacement for external function-calling data.

Limits

These two artifacts belong to one checkpoint investigation, not proof of two different training recipes. The precise parser-fix execution provenance is incomplete; unlike later strict rescoring, the filename alone does not establish that inference was not rerun.

Older synthetic partitioning and LoRA-initialization issues limit retrospective claims of clean controlled comparisons.

Supporting Files

- qwen2_5_1_5b_lora_baymax_control_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Control Data With Contrast Pairs

Experiment index

Question and Context

Could related ACT and non-ACT examples teach the model the difference between a complete request and one that should not be acted on?

This was an early contrast-pair trial, before the later anchored matched-group experiment.

Setup

The training source remained BAYMAX synthetic control data. Contrast pairs were added to make behavioral alternatives more explicit.

The historical result does not pin the selected pairs, effective token distribution, checkpoint hash or exact prompt. Do not substitute today's generator behavior for the missing historical recipe.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Control contrast pairs	50	12/50 (0.240)	0.530	0.767	0.111	0.071

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Control contrast pairs: 2026-08-22 UTC, 50 cases.

Interpretation

The run completed 12/50 tasks. Clarification and refusal keyword scores were still low. This supported further investigation of boundary training, but did not demonstrate that contrast pairs generalize reliably.

Limits

Comparisons with the previous control evaluation are historical, not a verified single-variable ablation. The later matched experiment changes group selection and coverage much more explicitly; it is not a repeat of this trial.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Control Data With Targeted Refusals

Experiment index

Question and Context

Could explicit refusal examples reduce unsupported actions when early control training still rarely refused?

Setup

Refusal-focused synthetic examples were added to the control-data path. A full 50-case evaluation and later seven-case prompt diagnostics survive.

The seven-case outputs are refusal-only diagnostics, not 50-case benchmarks. Some of those tests changed the inference prompt or parser; their numbers must not be attributed solely to training.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Targeted refusals	50	12/50 (0.240)	0.500	0.759	0.111	0.143
Targeted refusals / prompt-only	7	0/7 (0.000)	0.429	1.000	n/a	0.000
Control / policy prompt	7	0/7 (0.000)	0.429	1.000	n/a	0.071
Control / policy prompt / early parser fix	7	1/7 (0.143)	0.429	1.000	n/a	0.143

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Targeted refusals: 2026-08-22 UTC, 50 cases.
- Targeted refusals / prompt-only: 2026-09-15 UTC, 7 cases.
- Control / policy prompt: 2026-09-15 UTC, 7 cases.
- Control / policy prompt / early parser fix: 2026-09-15 UTC, 7 cases.

Interpretation

The full run completed 12/50 tasks and had refusal wording accuracy 0.143. Most seven-case diagnostics still recorded no strict task success; the later policy-prompt/parser-fix diagnostic recorded one.

Giving the model refusal examples did not by itself resolve boundary routing or explanation quality.

Limits

The saved files do not fully recover the authored-data changes or repeat counts. Single-class subset rates cannot establish balanced ACT/CLARIFY/REFUSE performance. Parser and prompt changes are additional variables.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Original External and BAYMAX Training Mix

Experiment index

Question and Context

Could generic xLAM tool-use training and BAYMAX-specific boundary examples complement each other?

This is the early "old mix" reference. It is not the later format-aligned or anchored reference.

Setup

Training combined external xLAM examples with authored BAYMAX control examples. The source recipes evolved, and the result files do not preserve a complete immutable account of the exact selected rows or sampling weights.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Original mixture	50	12/50 (0.240)	0.420	0.829	0.028	0.000
Original mixture / early parser fix	50	12/50 (0.240)	0.420	0.829	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Original mixture: 2026-08-22 UTC, 50 cases.
- Original mixture / early parser fix: 2026-08-22 UTC, 50 cases.

Interpretation

Both historical evaluations recorded 12/50 task successes, clarification accuracy 0.028 and refusal accuracy zero. Mixing the sources did not automatically teach abstention.

The initial source formats were not fully matched to evaluation: plain user queries and generic function arrays were being compared with structured time/state requests and BAYMAX control decisions. The later format-alignment experiment addresses that distribution shift.

Limits

Identical aggregate numbers in the early-parser-fix file are not proof that every output was identical. These artifacts predate later parser and split-integrity protections.

Use the linked original token audit for the distribution observed in its own 5,000-row artifact, not as a description of every early mixed checkpoint.

Supporting Files

- original-mix-token-audit-5000-and-boundary-failures.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Expanded Control Generator: Early Mixed Trial

Experiment index

Question and Context

Would broader authored clarification, refusal and confirmation examples improve the mixed model?

Setup

This is the synthetic-v7 mixed-data trial. The artifact is named "smoke", but its evaluation contains all 50 v1 scenarios. Evaluation size and training size are different facts.

The user discussed a 5,000-row trial during this period. The surviving result does not independently pin that exposure, so the note does not treat the filename as a training manifest.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Expanded control mix	50	15/50 (0.300)	0.500	0.865	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Expanded control mix: 2026-09-15 UTC, 50 cases.

Interpretation

Task success reached 15/50, but clarification and refusal keyword metrics were both zero. Better productive task completion did not establish better boundary judgment.

Limits

No frozen effective class/token counts or exact recipe are stored in the result. Do not compare this 50-case rate directly with later 150-case rates. The generator version is a historical label, not an immutable checkpoint identity.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Weighted Control Examples: Early Oversampling Trial

Experiment index

Question and Context

Would increasing the exposure of clarification and refusal examples make the model use those decisions?

Setup

A weighted/oversampled version of the synthetic-v7 mixture was trained. Weighting changes how often an existing example appears; it does not add semantic diversity.

The precise repeat settings and effective training prefix are not pinned in the saved evaluation. Later artifact audits were added because intended source weights did not reliably describe the data surviving a cap.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Weighted control mix	50	11/50 (0.220)	0.540	0.809	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Weighted control mix: 2026-09-15 UTC, 50 cases.

Interpretation

The run completed 11/50 tasks, versus 15/50 in the early unweighted v7 artifact. Clarification remained near zero and refusal remained zero.

This particular weighting attempt did not establish improvement. It does not prove all forms of weighting are ineffective.

Limits

The comparison is historical and lacks sufficient provenance for a clean causal attribution. Repetition can crowd out unique data, but these metrics alone cannot demonstrate memorization.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Clarification and Refusal as Explicit Meta-Tools

Experiment index

Question and Context

Was the model failing to abstain because clarification and refusal were not presented as recognizable actions?

Setup

The synthetic-v8 weighted path represented non-productive decisions as explicit calls:

- request_clarification with arguments.question.
- refuse_request with arguments.reason.

The local runner converts those calls into a BAYMAX response decision and message, rather than executing them as external mutations.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Meta-tool mixture	50	12/50 (0.240)	0.540	0.840	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Meta-tool mixture: 2026-09-15 UTC, 50 cases.

Interpretation

The run completed 12/50 tasks, but both wording metrics were zero. Adding action names alone did not resolve the problem.

The next important investigation was whether the model saw the same prompt, tool definitions and structured request wrapper during training and evaluation.

Limits

A zero keyword metric does not alone show that every model decision was ACT. Explicit routing metadata was not available in this artifact. The exact oversampling recipe is also not preserved.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Aligning Training and Evaluation Formats

Experiment index

Question and Context

Was a training/evaluation format mismatch preventing the model from using clarification and refusal?

Generic examples taught a plain request plus tools. BAYMAX evaluation supplied a structured request with current_time, initial_state and request, and injected two meta-tools. Those are meaningfully different input distributions.

Change

Align the structured wrapper, shared meta-tool definitions, names, ordering and system-prompt expectations between training and local evaluation. Non-ACT targets remain actual JSON calls, not prose-only answers.

This checkpoint became the historical "aligned" reference. It is different from the later target-compatible anchored two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Format-aligned	50	14/50 (0.280)	0.580	0.853	0.056	0.071
Format-aligned / historical settings	150	42/150 (0.280)	0.473	0.771	0.083	0.019
Format-aligned / matched training prompt	150	40/150 (0.267)	0.453	0.771	0.025	0.017
Format-aligned	30	5/30 (0.167)	0.517	0.817	0.100	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Format-aligned: 2026-09-16 UTC, 50 cases.
- Format-aligned / historical settings: 2026-09-26 UTC, 150 cases.
- Format-aligned / matched training prompt: 2026-09-26 UTC, 150 cases.
- Format-aligned: 2026-09-27 UTC, 30 cases.

Interpretation

On v1-50, success was 14/50 and clarification/refusal wording scores became nonzero. On the historical combined-150 evaluation, success was 42/150.

When tested with the recorded training prompt on combined-150, success was 40/150; template-shift success was 5/30. Alignment helped the engineering contract, but generalization remained weak.

Limits

The evaluations differ in suite, prompt and policy-era scoring; they are not repeated measurements of one fixed benchmark setup. Historical results do not fully pin selected training rows or base-model revision.

A format correction is necessary infrastructure, not proof that the decision policy has been learned robustly.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Extra Clarification and Refusal Repeats

Experiment index

Question and Context

The aligned model still acted prematurely. Could greater exposure to boundary examples help without changing LoRA settings?

Change

The named recipe added six extra clarification repeats and twelve extra refusal repeats. Repeats increase exposure to existing examples; they are not six and twelve newly authored unique requests.

Post-mix token audits were retained in two snapshots. Treat each snapshot as evidence about its reported artifact, not automatic provenance for the evaluated checkpoint.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Extra boundary repeats	50	8/50 (0.160)	0.420	0.689	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Extra boundary repeats: 2026-09-16 UTC, 50 cases.

Interpretation

Success was 8/50, below the historical aligned 14/50. Clarification wording accuracy remained low and refusal was zero.

This trial did not support continuing to increase repetition blindly. It motivated examining what actually reached training after shuffling, capping and label filtering.

Limits

The saved evaluation does not pin every training knob or the effective cap. Exact snapshot-to-checkpoint identity is incomplete. No memorization diagnosis follows from repetition counts alone.

Supporting Files

- clarification-extra6-refusal-extra12-token-audit-5000-snapshot1.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot2.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Legacy Boundary-Weighting Trial

Experiment index

Question and Context

This is a separate early weighting result that could otherwise be confused with the extra-six/extra-twelve run.

Recoverable Setup

The stored filename identifies a boundary-weighting trial historically called "legacy13". The evaluation contains 50 scenarios. Its exact repeat factors and training recipe are not established by that label.

Two token-audit snapshots exist. They should be inspected as separate snapshots; neither should be silently assumed to describe all historical weighting trials.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Legacy weighting	50	12/50 (0.240)	0.560	0.783	0.111	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Legacy weighting: 2026-09-16 UTC, 50 cases.

Interpretation

Success was 12/50, clarification wording accuracy 0.111 and refusal zero. It did not solve refusal behavior.

Limits

The recipe is not sufficiently pinned for exact reproduction or a one-variable comparison with the other weighted runs. This note deliberately preserves that uncertainty rather than assigning today's configuration to an old result.

Supporting Files

- boundary-weighting-legacy13-token-audit-5000-snapshot1.json

- boundary-weighting-legacy13-token-audit-5000-snapshot2.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Fixed Unique Boundary Pool at 1,000 Rows

Experiment index

Question and Context

Global shuffling and truncation could retain repeated boundary examples while discarding newly authored unique examples. Could deliberate unique-first sampling produce a better training artifact?

Change

Preserve a fixed BAYMAX contribution and fill the rest of the cap with unique xLAM rows. The observed 1,000-row raw cap contained 150 BAYMAX examples, including 60 unique refusals and 45 unique clarifications.

The owner's post-tokenization report retained 967 rows: 411 direct calls, 451 multi-tool sequences, 60 refusals and 45 clarifications. BAYMAX remained 150; xLAM retained 817. Exact duplicates were zero.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Boundary pool / 1,000 cap	50	12/50 (0.240)	0.560	0.746	0.167	0.143
Boundary pool / 1,000 cap / historical settings	150	37/150 (0.247)	0.513	0.738	0.083	0.056
Boundary pool / 1,000 cap / matched training prompt	150	37/150 (0.247)	0.493	0.738	0.050	0.086

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Boundary pool / 1,000 cap: 2026-09-16 UTC, 50 cases.
- Boundary pool / 1,000 cap / historical settings: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,000 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

The 150-case training-prompt evaluation completed 37/150 tasks. This sampler solved the exact-duplicate problem in the reported artifact; that is separate from proving strong policy transfer.

Limits

Earlier cap audits had different counts. The numbers above describe the corrected report, not every artifact historically called boundary-diverse.

The 50-case, historical-prompt and training-prompt evaluations are different measurement settings. Older split/initialization defects still limit causal claims.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Increasing the Cap to 1,500 Without Repeating Boundaries

Experiment index

Question and Context

Would more unique external training data help once the sampler stopped duplicating the fixed BAYMAX boundary pool?

Change

Increase the raw cap from 1,000 to 1,500, keeping BAYMAX at 150, including the same absolute quotas of 60 refusals and 45 clarifications. Extra capacity is filled primarily with unique xLAM rows.

The reported usable artifact had 1,447 rows: 657 direct, 685 multi-tool, 60 refusal and 45 clarification. Source counts were BAYMAX 150 and xLAM 1,297, with zero exact duplicates.

Refusal/clarification token shares fell as generic ACT exposure increased. A fixed row contribution does not mean a fixed class proportion.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Boundary pool / 1,500 cap	50	10/50 (0.200)	0.500	0.725	0.056	0.071
Boundary pool / 1,500 cap / historical settings	150	35/150 (0.233)	0.487	0.708	0.071	0.074
Boundary pool / 1,500 cap / matched training prompt	150	34/150 (0.227)	0.473	0.708	0.013	0.103

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Boundary pool / 1,500 cap: 2026-09-26 UTC, 50 cases.
- Boundary pool / 1,500 cap / historical settings: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,500 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

Under the matched training prompt, success was 34/150 versus 37/150 for the 1,000-row candidate. More rows did not yield a broad improvement in this run.

Limits

Changing the cap also changes optimizer exposure and class proportions. This is not an isolated test of semantic diversity alone. Single runs do not establish that a larger dataset is generally harmful.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

First Policy Prompt: Evaluation Without Retraining

Experiment index

Question and Context

Would explicitly describing ACT, CLARIFY and REFUSE improve the existing models at inference time?

Change

Evaluate bare Qwen and the existing aligned adapter with the first decision-policy prompt. The aligned checkpoint was not retrained with this new prompt.

The intended policy gives refusal precedence for unsafe/unavailable capabilities, clarification for recoverable missing details or required confirmation, and ACT for supported complete requests.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / policy prompt v1	150	37/150 (0.247)	0.537	0.827	0.013	0.034
Format-aligned / policy prompt v1	150	26/150 (0.173)	0.433	0.653	0.100	0.069

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / policy prompt v1: 2026-09-26 UTC, 150 cases.
- Format-aligned / policy prompt v1: 2026-09-26 UTC, 150 cases.

Interpretation

The new prompt measured 37/150 successes for base Qwen and 26/150 for aligned. That contrast raised a prompt-distribution-shift concern: a checkpoint trained with one prompt may behave differently under a new policy prompt.

Limits

This is a prompt-only experiment, not evidence of policy-aligned fine-tuning. Historical policy-era fixtures/scoring and previous prompt evaluations also differ, so not every historical comparison is a controlled prompt ablation.

The saved failure summaries diagnose the outputs of these runs. They are not labels to use for fixture-specific training.

Supporting Files

- qwen-base-policy-prompt-v1-failure-summary.txt
- format-aligned-policy-prompt-v1-failure-summary.txt

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Expanded Policy Prompt: Evaluation Without Retraining

Experiment index

Question and Context

Could a more explicit policy explanation correct the shortcomings of the first inference prompt?

Change

Use an expanded policy prompt on the same base and aligned model candidates. No newly trained policy-prompt adapter is established by these artifacts.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / policy prompt v2	150	5/150 (0.033)	0.480	0.521	0.113	0.052
Format-aligned / policy prompt v2	150	6/150 (0.040)	0.447	0.492	0.237	0.190

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / policy prompt v2: 2026-09-26 UTC, 150 cases.
- Format-aligned / policy prompt v2: 2026-09-26 UTC, 150 cases.

Interpretation

Task success dropped to 5/150 for base Qwen and 6/150 for aligned, while clarification/refusal wording scores rose. More boundary-language matches did not imply better overall behavior.

A longer or clearer human-readable instruction is not automatically a better prompt for a small checkpoint trained under a different format.

Limits

These are single inference evaluations. They do not prove prompt length itself caused the regression, nor that Qwen cannot learn the policy. Do not relax evaluation targets to turn the wording gains into successes.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Returning to the Recorded Training Prompt

Experiment index

Question and Context

Were changes in inference instructions making checkpoint comparisons misleading?

Change

Evaluate base Qwen, aligned and both boundary-diverse candidates with the same recorded training prompt. Its SHA256 is:

`a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468`

This did not retrain the adapters. It restored a matched inference prompt for the comparison.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / matched training prompt	150	32/150 (0.213)	0.493	0.791	0.025	0.034
Format-aligned / matched training prompt	150	40/150 (0.267)	0.453	0.771	0.025	0.017
Boundary pool / 1,000 cap / matched training prompt	150	37/150 (0.247)	0.493	0.738	0.050	0.086
Boundary pool / 1,500 cap / matched training prompt	150	34/150 (0.227)	0.473	0.708	0.013	0.103

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / matched training prompt: 2026-09-26 UTC, 150 cases.
- Format-aligned / matched training prompt: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,000 cap / matched training prompt: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,500 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

Aligned recovered to 40/150 successes, compared with 26/150 under policy prompt v1 and 6/150 under v2. Prompt compatibility was a real measurement concern.

Neither boundary-diverse candidate dominated aligned: additional boundary exposure improved some abstention metrics while reducing other task outcomes.

Limits

This comparison does not prove that the old training prompt is the best possible policy. It demonstrates why prompt changes must be tracked independently from training changes.

The historic suite policy and parser later changed. Keep these artifacts as historical evidence, not directly interchangeable with today's strict-parser results.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

One Hundred ACT, CLARIFY and REFUSE Examples

Experiment index

Question and Context

Could a larger unique BAYMAX contribution improve on the fixed 150-example pool without oversampling exact duplicates?

Change

The 1,000-row mixture targeted 300 unique BAYMAX examples: 100 productive ACT, 100 CLARIFY and 100 REFUSE, plus 700 generic xLAM productive examples.

"Balanced BAYMAX" is not "balanced full dataset": the xLAM portion still adds ACT targets, including multi-tool sequences.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX	150	43/150 (0.287)	0.567	0.761	0.050	0.172
100/100/100 BAYMAX	30	6/30 (0.200)	0.600	0.800	0.200	0.150
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
100/100/100 BAYMAX / strict replay	30	6/30 (0.200)	0.533	0.733	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113
100/100/100 BAYMAX / strict replay (30 cases)	0.435	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX: 2026-09-27 UTC, 150 cases.
- 100/100/100 BAYMAX: 2026-09-27 UTC, 30 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical full evaluation recorded 43/150 successes and the shift suite 6/30. Strict-parser replay reduced the full count to 42/150 and some argument/wording metrics.

This was a useful improvement over some early mixes, but the shift suite still exposed poor generalization and output-contract errors.

Limits

The historical and strict-replay rows are not different trained models. Use strict replay when comparing with the other replayed candidates. Old synthetic split and LoRA seed issues were not fixed by rescoring.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Early Balanced BAYMAX Data at Two Epochs

Experiment index

Question and Context

Would longer training help the early 100/100/100 BAYMAX mixture learn its policy?

Change

An early two-epoch SFT run used the same general ACR mixture family. This is not the later cleaned, anchored two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX / two epochs	150	46/150 (0.307)	0.570	0.791	0.075	0.138
100/100/100 BAYMAX / two epochs	30	5/30 (0.167)	0.467	0.728	0.250	0.400

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- 100/100/100 BAYMAX / two epochs: 2026-09-27 UTC, 150 cases.
- 100/100/100 BAYMAX / two epochs: 2026-09-27 UTC, 30 cases.

Interpretation

The historical full-suite count increased from 43 to 46 successes relative to the one-epoch ACR artifact. The template-shift count fell from six to five.

That result did not establish robust generalization from the extra epoch.

Limits

These runs predate later strict-parser and data-integrity corrections. No strict-rescore artifact for this two-epoch candidate is present in the catalog. The current two-epoch config is not sufficient evidence of an immutable historical data identity.

Supporting Files

- qwen2_5_1_5b_lora_baymax_mix_2epoch_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Schema-Diverse Control Examples

Experiment index

Question and Context

Was the model recognizing authoring templates without learning the tools' real argument schemas?

Change

Diversify synthetic productive examples and schema usage while keeping the broad 100 ACT / 100 CLARIFY / 100 REFUSE BAYMAX contribution.

The goal was transfer across wording and arguments, not rewriting evaluation cases or accepting malformed outputs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Schema-diverse	150	42/150 (0.280)	0.560	0.752	0.075	0.190
Schema-diverse	30	5/30 (0.167)	0.533	0.733	0.300	0.200
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Schema-diverse / strict replay	30	5/30 (0.167)	0.500	0.700	0.300	0.150

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Schema-diverse / strict replay (150 cases)	0.395	0.133
Schema-diverse / strict replay (30 cases)	0.323	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Schema-diverse: 2026-09-27 UTC, 150 cases.

- Schema-diverse: 2026-09-27 UTC, 30 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Historical full-suite success was 42/150; strict replay measured 40/150. Shift success was 5/30.

Higher boundary wording scores than some ACT-heavy variants did not make this checkpoint a universally better model. Correct routing, valid output and correct arguments remain separate capabilities.

Limits

A template-shift argument mismatch can be a genuine schema error, not a formatting false negative. Scorer changes require independent justification.

The old split/initialization caveats remain. Today's generator implementation cannot by itself recover the exact historical selection.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Increasing Productive BAYMAX Examples to Two Hundred

Experiment index

Question and Context

Could more domain-specific productive examples recover task completion without adding more clarification/refusal rows?

Change

The historical ACT-balanced mixture selected 400 BAYMAX rows: 200 ACT, 100 CLARIFY and 100 REFUSE. The remaining 600 raw slots came from xLAM.

This is the old ACT-balanced model, not generator14 ACT200 or the cleaned anchored checkpoint. The similar class counts do not make the data revisions identical.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT200 / legacy metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / decision metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / legacy metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT200 / strict replay	30	6/30 (0.200)	0.500	0.794	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT200 / decision metrics (150 cases)	0.387	0.000

Evaluation	Macro-F1	Invalid-output rate
Historical ACT200 / decision metrics (30 cases)	0.397	0.000
Historical ACT200 / strict replay (150 cases)	0.349	0.120
Historical ACT200 / strict replay (30 cases)	0.368	0.067

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical evaluation recorded 47/150 successes; strict replay recorded 46/150. Template-shift success remained 6/30.

The increase in productive exposure improved some established-suite outcomes but also increased premature actions. It was not a complete improvement in decision policy.

Limits

The "decision-metrics" evaluation added explicit decision reporting but did not create a new trained model. The "legacy-metrics" file remains preserved.

Strict replay fixes measurement only, not the old training partitions or seed initialization. Later ACT200 runs are not single-variable comparisons with this checkpoint.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples

Experiment index

Question and Context

Could a midpoint between 100 and 200 authored ACT examples retain productive accuracy without the same premature-action pressure?

Change

The historical ACT-midpoint candidate reduced the BAYMAX productive quota to 150 while retaining the boundary-training approach. It was evaluated with decision metadata and later strict replay.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT150 / decision metrics	150	46/150 (0.307)	0.520	0.804	0.025	0.052
Historical ACT150 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.050
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052
Historical ACT150 / strict replay	30	6/30 (0.200)	0.467	0.761	0.200	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / decision metrics (150 cases)	0.330	0.000
Historical ACT150 / decision metrics (30 cases)	0.343	0.000
Historical ACT150 / strict replay (150 cases)	0.308	0.107
Historical ACT150 / strict replay (30 cases)	0.321	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT150 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT150 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Full-suite success was 46/150 before and after replay; shift success was 6/30. Other metrics changed under strict parsing even though task-success totals stayed the same.

The result did not establish a clean win over ACT-balanced.

Limits

Matching success totals does not imply matching outputs or error types. The surviving historical result is not a complete frozen training manifest. Earlier seed/split defects still apply.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Content-Group Splits With Seventy Percent ACT

Experiment index

Question and Context

Earlier training partitions could share duplicated content or related contrast examples. New content-group splitting protected train/validation/test separation.

The owner also tried a more boundary-heavy mixture. Therefore this was not only a split-fix experiment.

Setup

- Generator12 synthetic data and content-grouped external data.
- Raw training cap 1,000: 150 BAYMAX ACT, 150 CLARIFY, 150 REFUSE and 550 xLAM ACT.
- Overall raw labels: 700 ACT / 150 CLARIFY / 150 REFUSE.
- Reported usable train artifact: 983 rows; direct 366, multi 317, refusal 150, clarification 150; BAYMAX 450, xLAM 533; zero exact duplicates.
- The 200-row validation prefix had almost no boundary data: one refusal and no clarification survived.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Content-group 70/30	150	26/150 (0.173)	0.427	0.579	0.113	0.207

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Content-group 70/30 (150 cases)	0.346	0.147

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Content-group 70/30: 2026-10-05 UTC, 150 cases.

Interpretation

The run completed 26/150 tasks. It improved some boundary wording metrics relative to ACT-heavy history, but productive performance was much worse.

Fixing leakage is essential for trustworthy methodology; it does not guarantee higher benchmark scores. This run cannot isolate whether splitting, composition or other data revisions explain the regression.

Limits

Validation loss on this mostly-ACT prefix was a poor monitor of boundary behavior. This generator predates subsequent adapter and policy corrections.

Supporting Files

- qwen2_5_1_5b_lora_content_groups_70_30_v12.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Corrected Policy, Credential Filtering and Representative Validation

Experiment index

Question and Context

The synthetic source had adapter-state defects and unresolved policy contradictions. External examples sometimes required model-emitted credentials. The small validation cap barely represented boundary decisions.

Change

Use corrected generator14 BAYMAX examples and credential-filtered xLAM. Resolved policy includes refusing indiscriminate outreach, not substituting drafts for unsupported sends, and allowing drafts composed from supplied facts.

External credential filtering excluded 618 rows from the original 60,000. Source data was preserved in a separate artifact.

Keep the boundary-heavy raw training recipe: 150 BAYMAX ACT / 150 CLARIFY / 150 REFUSE plus 550 xLAM ACT. A representative validation prefix selects 30 examples per BAYMAX decision plus 110 xLAM, from validation only.

Actual Artifact

Reported train: 976 usable rows, 374 direct, 302 multi-tool, 150 refusal and 150 clarification; BAYMAX 450 and xLAM 526; no exact duplicates.

Reported representative validation: 194 usable rows, 77 direct, 57 multi-tool, 30 refusal and 30 clarification; BAYMAX 90 and xLAM 104.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Corrected 70/30	150	25/150 (0.167)	0.433	0.582	0.125	0.190

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Corrected 70/30 (150 cases)	0.372	0.113

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Corrected 70/30: 2026-10-05 UTC, 150 cases.

Interpretation

Success was 25/150. Relative to generator12's 26/150, there was no clear overall improvement. Necessary correctness and security fixes should not be described as an optimization win.

Representative validation improved what loss could monitor. The trainer did not use early stopping or best-checkpoint selection, so changing this validation slice alone did not directly optimize the final weights.

Limits

Several source and validation changes occurred together. Validation losses from the sparse and representative prefixes are not directly comparable. Credential filtering did not remove all target-contract problems.

Supporting Files

- qwen2_5_1_5b_lora_auth_filtered_v14_val200.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Corrected Policy With Two Hundred Productive Examples

Experiment index

Question and Context

Would the productive-heavy 200/100/100 BAYMAX recipe work better with the corrected policy and credential-filtered external source?

Change

Use the same source family as the previous boundary-heavy run, but cap BAYMAX at 400: 200 ACT, 100 CLARIFY and 100 REFUSE. Fill 600 raw slots with unique xLAM ACT examples.

Keep a representative 200-row validation cap, the base model, one epoch, seed, prompt and LoRA settings. This is the "original ACT200" control used by subsequent cleaning comparisons.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Corrected ACT200	150	38/150 (0.253)	0.473	0.675	0.062	0.121

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Corrected ACT200 (150 cases)	0.396	0.133

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Corrected ACT200: 2026-10-05 UTC, 150 cases.

Interpretation

Success increased from 25/150 to 38/150; macro-F1 was 0.396. That recovered productive performance but reduced some boundary wording metrics.

The comparison supports a mixture tradeoff, not a claim that more ACT always helps.

Limits

It is not the same source as historical ACT-balanced, even though the broad quotas match. The measured 20 invalid outputs also motivated validating training targets separately from changing decision proportions.

Supporting Files

- qwen2_5_1_5b_lora_auth_filtered_v14_act200_val200.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Target-Contract Cleaning With Resampled External Rows

Experiment index

Question and Context

The new strict training gate found target/schema failures in xLAM. Would excluding incompatible examples help?

Change

Build a target-compatible external source: 57,611 accepted rows after 618 credential exclusions and 1,771 additional target/credential exclusions, for 2,389 total skipped rows.

Retain the generator14 ACT200 recipe, but rebuild the mixed sampler. Removing rows changes which xLAM examples survive the cap; this is the cleaned-resampled candidate.

Actual Artifact

Raw cap 1,000; usable train 977: direct 447, multi 330, refusal 100, clarification 100. BAYMAX 400, xLAM 577, zero exact duplicates.

Usable validation 196: direct 72, multi 64, refusal 30, clarification 30.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Cleaned / resampled	150	47/150 (0.313)	0.493	0.761	0.013	0.086

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Cleaned / resampled (150 cases)	0.354	0.073

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Cleaned / resampled (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	59	15	2	5
CLARIFY	28	3	5	4
REFUSE	17	2	7	3

Result Provenance

- Cleaned / resampled: 2026-10-06 UTC, 150 cases.

Interpretation

Success reached 47/150, versus 38/150 for original ACT200. Invalid outputs fell from 20 to 11, but routing macro-F1 fell from 0.396 to 0.354.

The model completed more tasks while handling abstention worse. "Cleaning improved reasoning" would overstate the evidence.

Limits

A recorded content-hash comparison found only 7/600 shared xLAM rows and 301/400 shared BAYMAX rows versus the original raw cap. Even the unchanged BAYMAX source yielded different selected examples. This does not isolate removal of faulty examples. A full-pool exclusion rate of about three percent can still change a small capped artifact substantially. The anchored comparison was created to address that confound.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_val200.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Target-Contract Cleaning With Anchored External Rows

Experiment index

Question and Context

Did the earlier cleaned model change because of better targets, or because rebuilding the mixture selected different external examples?

Change

Anchor selection to original ACT200. Preserve all 400 capped BAYMAX rows and retain 582 of the original 600 xLAM rows; replace only the 18 excluded xLAM rows with compatible same-split rows.

Validation retained 90 BAYMAX and 107 xLAM rows, replacing three xLAM examples. Train/validation/test membership remains protected. The raw recipe stays 200 BAYMAX ACT / 100 CLARIFY / 100 REFUSE plus 600 xLAM.

Actual Artifact and Recipe

One fresh SFT epoch, base Qwen2.5-1.5B-Instruct, seed 1, LoRA rank 8/alpha16 on q_proj and v_proj, dropout 0.05. Learning rate 2e-4, cosine scheduler, warmup fraction 0.03.

Owner low-VRAM overrides: context 1,024, batch one, accumulation 16, raw train/eval caps 1,000/200.

Usable train: 974 rows, direct 435, multi 339, refusal 100, clarification 100; BAYMAX 400, xLAM 574, no exact duplicates. Validation retained 195.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / one epoch	150	39/150 (0.260)	0.500	0.708	0.050	0.155

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / one epoch (150 cases)	0.429	0.080

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / one epoch (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	50	22	3	6
CLARIFY	24	7	5	4
REFUSE	12	2	12	3

Result Provenance

- Anchored / one epoch: 2026-10-06 UTC, 150 cases.

Interpretation

Success was 39/150, macro-F1 0.429 and invalid outputs 12/150. Compared with resampling's 47 successes but 0.354 macro-F1, anchoring recovered more balanced routing but not its task total.

This became the data reference for the epoch-length trials.

Limits

Replacing 18 examples also changes token lengths and usable counts. This is a much tighter comparison, not perfectly identical exposure. Single-seed measurements do not establish a statistically reliable causal effect.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Anchored Data at Two Epochs: Selected Reference

Experiment index

Question and Context

Was one pass over the cleaned anchored mixture insufficient to learn its targets?

Controlled Change

Increase fresh SFT from one to two total epochs. Keep the selected data, prompt, seed, raw caps and LoRA settings fixed. A longer run also stretches the cosine schedule and warmup in steps, so this is not an extra pass at the exact previous per-step learning rates.

Actual Training Recipe

- Fresh base Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA; not a resumed adapter.
- LoRA rank 8, alpha 16, q_proj/v_proj, dropout 0.05, bias none.
- Learning rate 2e-4, cosine schedule, warmup fraction 0.03, seed 1.
- Owner overrides: context 1,024, train/eval batch one, accumulation 16, raw caps 1,000/200.
- Assistant-only SFT token cross-entropy; prompt/padding tokens do not provide target loss.
- Recorded training/inference system prompt hash a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.

The same usable artifact retained 974 train rows: 435 direct, 339 multi-tool, 100 clarification and 100 refusal. BAYMAX contributed 400 and xLAM 574. Validation retained 195. There were zero exact duplicates.

The trainer completed 122 optimizer steps.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / two epochs	150	44/150 (0.293)	0.600	0.771	0.075	0.172
Anchored / two epochs	30	5/30 (0.167)	0.433	0.694	0.250	0.250

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / two epochs (150 cases)	0.514	0.013
Anchored / two epochs (30 cases)	0.345	0.067

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

Decision Matrix: Anchored / two epochs (30 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	3	6	1	0
CLARIFY	5	2	3	0
REFUSE	3	0	5	2

Result Provenance

- Anchored / two epochs: 2026-10-06 UTC, 150 cases.
- Anchored / two epochs: 2026-10-07 UTC, 30 cases.

Interpretation

Combined success increased from 39 to 44/150, macro-F1 from 0.429 to 0.514, and invalid outputs fell from 12 to two. Premature productive decisions on boundary requests fell from 36 to 31.

This is the selected development reference because it balances routing and validity better than the later candidates, not because it is a solved assistant. Shift success was only 5/30, with eight premature ACT decisions on the twenty boundary cases.

Reproducibility

The linked frozen reference records data revision, config/prompt hashes, capped raw-input hashes, adapter identity and both evaluations. "Frozen" means identities recorded for drift detection, not permission-locked files.

Dataset revision: `329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c`.

Canonical training config hash: `ce9efa2db24a065dbdb4eb482c07a29a56bfbcbdce32495418eb54b92d985c40`.

Limits

Exact base-model revision was not recorded. Some inference settings come from owner commands rather than embedded result metadata. Results are single-seed, repeatedly inspected development evidence. An earlier bounded audit found near-duplicate training/evaluation wording; the benchmark is not an untouched final test.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2.json
- anchored-epochs2-reference.json
- anchored-epochs2-vs4-failure-audit.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Anchored Data at Four Epochs

Experiment index

Question and Context

Would substantially longer training improve the two-epoch reference, or simply strengthen the wrong decision patterns?

Controlled Change

Train a fresh adapter for four total epochs on the same anchored data. This was not continuing from the saved two-epoch adapter.

- Fresh base Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA; not a resumed adapter.
- LoRA rank 8, alpha 16, q_proj/v_proj, dropout 0.05, bias none.
- Learning rate 2e-4, cosine schedule, warmup fraction 0.03, seed 1.
- Owner overrides: context 1,024, train/eval batch one, accumulation 16, raw caps 1,000/200.
- Assistant-only SFT token cross-entropy; prompt/padding tokens do not provide target loss.
- Recorded training/inference system prompt hash a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.

The usable training/validation counts stayed 974/195, with the same broad class counts and no exact duplicates. The learning-rate schedule now spans four epochs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / four epochs	150	37/150 (0.247)	0.440	0.628	0.263	0.103
Anchored / four epochs	30	4/30 (0.133)	0.400	0.528	0.300	0.050

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / four epochs (150 cases)	0.372	0.087
Anchored / four epochs (30 cases)	0.348	0.200

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / four epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	39	36	1	5
CLARIFY	14	19	4	3
REFUSE	17	3	4	5

Decision Matrix: Anchored / four epochs (30 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	2	7	1	0
CLARIFY	1	8	1	0
REFUSE	3	0	1	6

Result Provenance

- Anchored / four epochs: 2026-10-07 UTC, 150 cases.
- Anchored / four epochs: 2026-10-07 UTC, 30 cases.

Interpretation

Compared with two epochs, combined task successes fell from 44 to 37, macro-F1 from 0.514 to 0.372, and invalid outputs increased from two to thirteen.

Correct CLARIFY routing rose from 11/40 to 19/40, but correct ACT fell from 57/81 to 39/81 and correct REFUSE from 16/29 to 4/29. Higher clarification wording scores therefore concealed a harmful routing tradeoff.

Shift success also fell, from five to four, and invalid outputs rose from two to six. Its macro-F1 was nearly unchanged, but the class-specific behavior changed substantially.

Why Lower Loss Was Not Enough

Validation token loss fell to 0.2364 from the one-epoch run's 0.3107. That objective rewards the supplied answer tokens; it does not directly optimize routing macro-F1, execution success or semantic explanations.

The loss reduction does not prove useful generalization, nor does the regression prove memorization. It does show that more epochs are not a reliable fix under this recipe.

Decision and Limits

Keep two epochs as the reference. The saved audit separates wrong decisions, wording-score mismatch, argument errors and invalid output without repairing results.

This was one seed, and changing epochs changes the schedule horizon. Do not generalize it to every learning rate, dataset or local model.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs4.json
- anchored-epochs2-vs4-failure-audit.md
- anchored-epochs2-vs4-failure-audit.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Matched Contrasts at Fixed Class Counts

Experiment index

Question and Context

Could tighter ACT/CLARIFY/REFUSE contrasts teach the decision boundary better than isolated examples, without changing broad class quantities?

A useful contrast changes missing information, state or capability while keeping the surrounding task similar. It should teach the relevant distinction, not a keyword shortcut.

Controlled Change

Start from the anchored two-epoch recipe. Preserve all 600 raw xLAM positions and validation/test files byte-for-byte. Replace 398 of 400 BAYMAX training rows; keep two identical.

Raw quantities remain 200 BAYMAX ACT / 100 CLARIFY / 100 REFUSE. Selection prioritized complete training-only contrast groups: 120 groups, comprising 73 triplets and 47 pairs. Held-out overlap checks remained enforced.

Usable rows stayed 974, including all 400 BAYMAX examples. No exact duplicates were present. Training settings and prompt matched the two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Matched contrasts / two epochs	150	39/150 (0.260)	0.480	0.698	0.087	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Matched contrasts / two epochs (150 cases)	0.383	0.053

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Matched contrasts / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	48	29	2	2
CLARIFY	20	14	6	0
REFUSE	16	2	5	6

Result Provenance

- Matched contrasts / two epochs: 2026-10-07 UTC, 150 cases.

Interpretation

Task successes fell from 44 to 39/150, macro-F1 from 0.514 to 0.383, and invalid outputs rose from two to eight. Correct refusal routing fell from 16/29 to 5/29.

The separate audit found real capability/routing errors as well as missing tool names and wrong meta-tool fields. This was not merely keyword scoring rejecting semantically correct refusals.

Important Confound

Broad class counts stayed fixed, but within-class coverage did not. Refusal examples from triplet origins increased from 33 to 73; standalone and capability-pair origins disappeared from the cap. Source-ID family count fell from 45 to 20.

Those are authoring lineages, not a complete semantic taxonomy. The plausible hypothesis is narrower coverage, not a demonstrated causal explanation.

Decision and Limits

Retain the original anchored two-epoch reference. This matched implementation regressed; it does not disprove contrastive training in general. A future contrast experiment would need to preserve semantic/domain coverage as well as label counts.

The earlier auth-filtered matched preflight was blocked by xLAM target failures; it was a different preparation, not a completed precursor checkpoint.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_matched_anchored_val200_epochs2.json
- matched-contrasts-epochs2-refusal-format-audit.md
- matched-contrasts-epochs2-refusal-format-audit.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Two-to-One Productive-to-Boundary SFT Ratio

Experiment index

Question and Context

Would a more boundary-heavy mixture help, following the owner's discussion of a 2:1 productive/non-productive ratio?

This is an empirical BAYMAX ratio test. It is not evidence that a paper's ratio transfers unchanged to this model and dataset.

Change

Keep the original anchored two-epoch BAYMAX pool, add 64 unique clarifications and 70 unique refusals, and replace 134 capped xLAM positions. Preserve validation/test bytes and other training settings.

Raw training cap: 666 ACT / 164 CLARIFY / 170 REFUSE, rather than the reference's 800 ACT / 100 CLARIFY / 100 REFUSE. ACT includes both xLAM and productive BAYMAX examples.

Actual Artifact

Class	Usable rows	Trainable tokens
Direct productive call	372	11,109
Multi-tool sequence	272	14,062
CLARIFY	164	4,593
REFUSE	170	5,022

978 usable rows: BAYMAX 534, xLAM 444; no exact duplicates. The effective ACT:boundary row ratio is 644:334, approximately 1.93:1. Boundary tokens are 27.6% of target tokens, not one-third.

The initially proposed 167/167 allocation failed unique-input coverage. The corrected quota preserved the boundary total without repeating examples or borrowing held-out data.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
2:1 SFT / two epochs	150	29/150 (0.193)	0.487	0.623	0.125	0.259

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
2:1 SFT / two epochs (150 cases)	0.477	0.053

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: 2:1 SFT / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	31	38	6	6
CLARIFY	14	17	8	1
REFUSE	8	1	19	1

Result Provenance

- 2:1 SFT / two epochs: 2026-10-09 UTC, 150 cases.

Interpretation

Success fell from 44/150 to 29/150 and macro-F1 from 0.514 to 0.477. Premature actions fell from 31 to 22, but correct ACT routing fell from 57/81 to 31/81; ACT-to-CLARIFY errors rose from 21 to 38.

The mixture improved some boundary recall and wording scores at a large productive-task cost. It was not a balanced win.

Limits

Changing the ratio also changes examples, source proportions, token exposure and optimizer steps. This does not isolate class weights alone. The checkpoint remains useful as an alternative starting point for the next DPO comparison.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_2to1_anchored_val200_epochs2.json
- prepare_sft_2to1.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

DPO From the Selected Two-Epoch SFT Reference

Experiment index

Question and Context

Could preferences between correct and incorrect decisions sharpen routing after SFT, without changing the model or evaluator?

DPO trains relative preference for a chosen answer over a rejected answer to the same prompt. It is not an LLM judge reviewing benchmark results.

Training Recipe

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.
- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

Preference Artifact

400 unique BAYMAX training anchors: 200 ACT, 100 CLARIFY, 100 REFUSE. xLAM rows were not used as preference anchors. There were 705 pairs because an anchor can have multiple distinct rejections.

Chosen over rejected	Pairs
ACT > CLARIFY	200
ACT > REFUSE	200
CLARIFY > ACT	50
CLARIFY > REFUSE	100
REFUSE > ACT	55
REFUSE > CLARIFY	100

50 clarification and 45 refusal anchors lacked a suitable productive rejection. Missing directions were reported rather than filled with arbitrary tool calls.

Actual artifact: 23,008 chosen completion tokens, 20,981 rejected tokens, longest sequence 633, zero truncated pairs. The Windows smoke established runnable training, not improved capability.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	705
Unique training prompts	400
Chosen completion tokens	23,008
Rejected completion tokens	20,981
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	150
REFUSE	155

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / original reference	150	53/150 (0.353)	0.533	0.856	0.013	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / original reference (150 cases)	0.376	0.033

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / original reference (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	72	5	1	3
CLARIFY	35	4	1	0
REFUSE	21	1	5	2

Result Provenance

- DPO / original reference: 2026-10-09 UTC, 150 cases.

Interpretation

Task successes increased from SFT's 44 to 53/150, but macro-F1 fell from 0.514 to 0.376. Premature ACT decisions on boundary cases rose from 31 to 56. Correct refusal routing fell from 16 to five, and correct clarification from eleven to four.

This run favored productive action and argument accuracy. It was a task-completion gain paired with a substantial boundary regression, not a general improvement.

Limits

Preference directions were incomplete and asymmetric. That is a plausible contributor, not proven causation. Pair counts are not unique-prompt counts. Results are single-seed and repeatedly evaluated.

The initial Windows failure was a TRL/Triton compatibility problem, resolved with the ordinary PyTorch path; it did not change preference labels or the DPO objective.

Supporting Files

- qwen2_5_1_5b_anchored_epochs2_v1.json
- dpo-run.json
- DPO_EXPERIMENT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

DPO From the Two-to-One SFT Model

Experiment index

Question and Context

Could the more cautious 2:1 SFT checkpoint resist the ACT bias seen after DPO on the original reference?

Controlled Change

Change the initial policy and matching frozen reference to the 2:1 SFT adapter. Keep the exact original 705 preference pairs, prompt, DPO settings and evaluator unchanged.

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.
- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

"2:1 DPO" means DPO started from a 2:1-trained SFT model. The preference dataset itself was not rebalanced to 2:1.

Provenance

The config separates preference provenance (reference_config) from policy initialization (initial_sft_config). Starting config and adapter hashes are pinned. Actual artifact retained 705 pairs / 400 prompts, longest sequence 633, zero truncation.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	705
Unique training prompts	400
Chosen completion tokens	23,008
Rejected completion tokens	20,981
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	150
REFUSE	155

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / 2:1 start	150	31/150 (0.207)	0.473	0.676	0.125	0.172

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / 2:1 start (150 cases)	0.453	0.047

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / 2:1 start (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	40	32	3	6
CLARIFY	21	18	1	0
REFUSE	13	5	10	1

Result Provenance

- DPO / 2:1 start: 2026-10-09 UTC, 150 cases.

Interpretation

Against 2:1 SFT, task successes rose from 29 to 31/150, while macro-F1 fell from 0.477 to 0.453. Premature ACT decisions increased from 22 to 34, and correct refusal routing fell from 19 to ten.

The starting checkpoint affected the outcome, but DPO again shifted toward action. This was not an overall improvement over the original two-epoch SFT reference.

Limits

Changing initialization also changes the frozen DPO reference distribution, necessarily. It is not a test of changing only the policy weights while holding reference probabilities fixed.

The same preference-coverage gaps remained. No additional preference examples were introduced in this experiment.

Supporting Files

- qwen2_5_1_5b_2to1_epochs2_v1.json
- dpo-run.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Completing Missing Productive Rejections for Boundary Decisions

Experiment index

Question and Context

Did the first preference set underteach "clarify instead of act" and "refuse instead of act" because those rejected alternatives were missing for many anchors?

Change

Start fresh from the original anchored two-epoch SFT adapter, not the previous DPO adapter or the 2:1 SFT model.

Add 95 explicit incorrect ACT answers to existing training prompts: 50 clarification and 45 refusal anchors. Negatives deliberately guess missing details, skip confirmation, call a known unavailable operation, or substitute an inappropriate productive action.

Training-only counterparts take priority. Authoring explanations stay in the review sidecar, not the model input. No evaluation fixture supplied a target.

Preference Artifact

Chosen over rejected	Before	After
ACT > CLARIFY	200	200
ACT > REFUSE	200	200
CLARIFY > ACT	50	100
CLARIFY > REFUSE	100	100
REFUSE > ACT	55	100
REFUSE > CLARIFY	100	100

800 pairs, still 400 unique prompts and the same 200/100/100 unique anchor counts. Actual artifact: 25,854 chosen and 24,541 rejected completion tokens, longest sequence 633, no truncated pairs.

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.

- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	800
Unique training prompts	400
Chosen completion tokens	25,854
Rejected completion tokens	24,541
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	200
REFUSE	200

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / completed ACT rejections	150	46/150 (0.307)	0.533	0.793	0.050	0.155

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / completed ACT rejections (150 cases)	0.482	0.047

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / completed ACT rejections (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	1	6
CLARIFY	30	9	1	0
REFUSE	15	2	11	1

Result Provenance

- DPO / completed ACT rejections: 2026-10-09 UTC, 150 cases.

Interpretation

Task successes were 46/150, macro-F1 0.482 and invalid outputs seven. Compared with first DPO's 53 successes / 0.376 macro-F1, this recovered some routing balance at a task-completion cost.

Premature ACT decisions fell from 56 to 45, but the original SFT reference had only 31. Original SFT still had higher macro-F1 (0.514) and fewer invalid outputs (two).

The gap repair helped relative to the first DPO candidate, but did not beat the original SFT on balanced routing.

Limits

Adding 95 pairs increases optimizer exposure at the same epoch count: this is not a step-matched ablation. Coverage is complete but directions are not equally weighted: ACT-over-boundary has 400 pairs, boundary-over-ACT 200.

No claim follows that DPO cannot work, that Qwen is inherently unsuitable, or that longer DPO would fix it. Those remain untested hypotheses.

Supporting Files

- qwen2_5_1_5b_anchored_epochs2_act_rejections_v2.json
- dpo-run.json
- dpo_act_rejections.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Untrained Qwen: What the Baseline Actually Measures

Experiment index

Question and Context

Does fine-tuning help compared with the same local model without an adapter?

This note records the bare Qwen reference. Hosted GPT and scripted results are different baselines: hosted inference changes the model, while scripted responses test the harness rather than learned behavior.

Setup

Base Qwen2.5-1.5B-Instruct, no LoRA adapter, evaluated through the local runner and fake tools.

Use the training-prompt-a314 artifact when comparing with similarly prompted historical adapters. The "historical" artifact predates some policy/scoring changes and is not interchangeable.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / historical settings	150	32/150 (0.213)	0.500	0.791	0.036	0.019
Bare Qwen / matched training prompt	150	32/150 (0.213)	0.493	0.791	0.025	0.034
Bare Qwen	30	7/30 (0.233)	0.483	0.947	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / historical settings: 2026-09-26 UTC, 150 cases.
- Bare Qwen / matched training prompt: 2026-09-26 UTC, 150 cases.
- Bare Qwen: 2026-09-27 UTC, 30 cases.

Interpretation

The historical combined suite measured 32/150 task successes. The matched-training-prompt evaluation also had 32/150, but some tool/wording metrics differed. Template-shift success was 7/30.

This establishes that "same task-success rate" can hide different behavior. It also explains why 50-, 150- and 30-case percentages should not be placed in one improvement curve.

Limits

These base results predate later strict-parser measurements. A clean claim that a modern checkpoint beats base needs a method-matched base artifact, not simply the largest improvement over any historical baseline.

Reported local API cost of zero does not include electricity, hardware or training cost.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Effective Token Distribution and Training-Cap Audit

Experiment index

Question and Context

Did nominal row weights reflect the learning exposure actually delivered to SFT?

Method

Report assistant-label tokens after mixing, split selection, deterministic shuffling, caps, truncation and removal of rows with no usable labels. Count sources and exact unique rows as well as behavioral labels.

This is a data measurement, not a model training experiment.

Saved 5,000-Row Token Snapshots

Snapshot	Direct rows / tokens	Multi rows / tokens	Refusal rows / tokens	Clarification rows / tokens
Original mix	2,285 / 63,148	2,497 / 121,608	121 / 2,953	97 / 2,380
Extra-six / extra-twelve	2,125 / 60,266	2,331 / 115,158	356 / 8,745	188 / 4,658
Legacy boundary weighting	2,086 / 58,321	2,187 / 107,640	378 / 9,293	349 / 8,612

The two saved snapshots within each weighted family report the same token counts. That alone does not establish identical checkpoint identity.

Cap Failure and Correction

At one stage the 1,000-row artifact retained 60 refusal rows but only 36 unique ones, and 40 clarification rows but only 27 unique ones. More authored examples were not necessarily surviving the cap.

Unique-first sampling corrected that. Increasing the raw cap then exposed another defect: at 1,500, refusal rose to 103 rows but only 60 unique, and clarification to 68 but only 45 unique. There were 67 exact duplicates overall.

The corrected fixed-contribution reports were:

Raw cap	Usable rows	BAYMAX	xLAM	Unique refusal	Unique clarification	Exact duplicates
1,000	967	150	817	60	45	0
1,500	1,447	150	1,297	60	45	0

Interpretation

Rows and tokens answer different questions. Multi-tool targets contribute more tokens per example; a substantial boundary row count can still be a small share of training loss.

The sampler fix preserves diversity and deterministic selection. It does not prove the model improved. The two boundary-pool training notes document those evaluations separately.

Limits

Earlier duplicate counts and corrected cap counts come from owner reports. No fresh tokenizer or full-corpus audit was run to write these notes. Exact duplicates are not the same as near-duplicates or template similarity.

Do not assign a 5,000-row snapshot to a particular checkpoint without confirming its provenance.

Supporting Files

- original-mix-token-audit-5000-and-boundary-failures.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot1.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot2.json
- boundary-weighting-legacy13-token-audit-5000-snapshot1.json
- boundary-weighting-legacy13-token-audit-5000-snapshot2.json
- report_sft_artifact.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Strict Parser Replay: Correcting Measurement, Not Training

Experiment index

Question and Context

Could permissive parsing execute a valid-looking part of an otherwise invalid model response, overstating benchmark performance?

Change and Method

Reparse saved raw generations with strict-complete-response-v1, replay fake tool execution, and write separate result files. No model inference or training was rerun.

Malformed calls, invalid meta-tool fields, conflicting decisions and trailing output reject the whole response. There is no accepted valid subset or automatic repair.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113
Schema-diverse / strict replay (150 cases)	0.395	0.133
Historical ACT200 / strict replay (150 cases)	0.349	0.120

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / strict replay (150 cases)	0.308	0.107

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.

Interpretation

Full-suite success counts changed from 43 to 42 for ACR, 42 to 40 for schema-diverse, 47 to 46 for ACT-balanced and stayed 46 for ACT-midpoint.

These differences correct the measuring instrument. They must not be counted as model regressions caused by new training.

Provenance and Limits

Replay artifacts retain source/scenario hashes and inherit the original generation cost/latency. That latency cannot measure replay performance or a new model-speed gain.

Old split leakage and unseeded LoRA initialization were not repaired by replay. Source filenames embedded in JSON can predate renaming; use FILE_RENAMES.json to locate their current paths.

The linked parser README documents rejected-output counts and the corrected synthetic split. Later routing metrics may be derived from saved decisions without changing these historical JSON files.

Supporting Files

- README.md
- FILE_RENAMES.json
- SCORING.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Training Data Contract and Policy Review

Experiment index

Question and Context

Were model failures partly being taught by inconsistent training examples rather than caused only by insufficient training?

This was a series of related source-quality audits, not a claim that all external data was bad or a completed model-training experiment.

BAYMAX Source Findings

An adapter audit checked 1,128 synthetic examples and executed all 388 productive target sequences in fresh fake state. It found:

- 188 clipboard states used a plain string instead of the adapter's state object.
- 17 calendar states lacked required event details.
- 43 clarification targets did not request all missing details.

The generator13 correction changed 248 distinct examples without turning invalid/ambiguous requests into valid ACT requests.

Owner-approved generator14 policy changes corrected 41 bulk-outreach explanations, one unavailable-send label, and 24 draft-composition triplet examples. Training targets were validated independently of benchmark outcomes.

External Source Findings and Filtering

A numeric audit scanned 60,000 rows and flagged 1,069 candidates. Flags required review; numeric coincidence or explicit rounding could be valid.

A deliberately enriched review of 100 examples found both metadata and target/prompt defects. It cannot estimate the whole dataset's defect rate.

The broad contract scan flagged 31,821 rows. That was not the removal list: defaults, ambiguous schemas and context-dependent findings are not equivalent to confirmed incompatible targets.

Credential filtering excluded 618 rows. A narrower target-compatibility audit on the remaining source proposed 1,771 additional exclusions, leaving 57,611 accepted examples. Retained source bytes/order/splits and reviewed exclusion identities were preserved.

Blocked Experiment

The first auth-filtered matched-data preparation failed whole-dataset validation and a capped training preflight reported 30 target issues, including duplicate tools, wrong types and unverifiable types.

No successful checkpoint/result is evidenced for that blocked preparation. The owner filtered the incompatible external targets before running subsequent cleaned experiments.

Interpretation

Mechanical validity and policy correctness are different checks. Passing a schema does not prove an answer is grounded, necessary or appropriate. Conversely, every broad warning is not grounds for dropping a row.

These corrections made training evidence more trustworthy. The resampled and anchored cleaning notes record the actual model outcomes; improvement must come from those results, not the audit counts.

Limits

Semantic quality remains incompletely audited, and no LLM judge was introduced. Evaluation cases were not used to rewrite source targets. The local review artifacts are not required to be committed or embedded in these notes.

Supporting Files

- TRAINING_DATA_AUDIT.md
- qwen2_5_1_5b_lora_auth_filtered_v14_matched_act200_val200.json
- audit_sft_numeric_arguments.py
- validate_sft_dataset.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Decision Metrics Separate From Explanation Wording

Experiment index

Question and Context

A valid refusal could choose the right decision but match only one required reason keyword. Were keyword scores being mistaken for routing accuracy?

Measurement Change

Preserve selected act/clarify/refuse in AgentResponse.decision_type, independently of the displayed clarification question or refusal reason.

Keep existing keyword and task-success rules intact. Add decision accuracy and decision coverage; later reporting derives macro-F1, invalid-output rate and the confusion matrix from saved decisions.

Results and Meaning

The historical ACT-balanced legacy and decision-metric evaluations both recorded 47/150 task successes, tool accuracy 0.553 and argument accuracy 0.805. On template shift both recorded 6/30 successes.

The reporting change did not train a better model. It revealed a different aspect of the same task: correct routing can coexist with an incomplete or false explanation.

Reading the Metrics

- Macro-F1 weights ACT, CLARIFY and REFUSE equally, instead of rewarding the majority class alone.
- Conditional decision accuracy excludes unknown decisions; always read it with coverage.
- Invalid outputs count as missed predictions for their expected class in macro-F1.
- Refusal/clarification accuracy measures the required message keywords, not just the selected label.
- Task success requires the broader scenario checks, not merely a correct decision.
- Hallucination rate currently measures unavailable tool names, not every invented fact.

Limits

An invalid-output rate is not the rate of every runtime/schema error. A parser-valid call can still have wrong tool arguments or fail execution.

Core-agent runtime outcome versus selected decision is a separate concern documented in the bug report. Do not assume a final refusal message always means the model chose REFUSE. This note records measurement semantics, not an assertion that every runner's runtime paths have been fixed.

Supporting Files

- SCORING.md
- BUG_REPORT.md
- routing_metrics.py
- act-balanced-train1k-combined-150-legacy-metrics.json
- act-balanced-train1k-combined-150-decision-metrics.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.

Clarification Output Contract: Target Audit and Failure Diagnostics

Experiment index

Question

Experiment C sometimes calls `request_clarification` with `reason`, although the shared contract requires `question`. Is this incorrect training supervision or an invalid model output? Can we distinguish it without repairing benchmark answers?

Training Targets Checked

The audit read the same capped raw prefixes selected by C, using the existing `load_sft_jsonl` and `validate_training_targets` functions. No model or tokenizer was loaded and no training data was changed.

Split	Raw targets checked	Clarification targets	Distinct question strings	Target-contract errors
Train	1,000	100	65	0
Validation	200	30	22	0

All selected clarification targets use a nonempty `question` field, not `reason`. Supplied meta-tool schemas match the shared definitions. Distinct target wording was inspected: it requests a time, duration, recipient, text, task/event identity, conflict resolution or confirmation. Examples include "How many minutes should the consultation last?" and "Please confirm you want me to send this resignation message." A confirmation request need not end with a question mark.

Repeated question strings do not imply duplicate complete training rows. This check covers raw capped targets, not newly measured post-tokenization artifacts, and does not prove semantic correctness against every example's entire world state. No automatic semantic question-quality grader was added.

Recorded C Output Failures

Raw output diagnostic	Count
Attempted clarification with invalid arguments	10
Other malformed calls	5
Unknown decision from an empty array	1

The ten clarification calls supplied `reason` instead of `question`. Six were on expected CLARIFY cases, two on ACT cases and two on REFUSE cases. Thus the diagnostic identifies an attempted clarification, not necessarily correct routing. The remaining malformed calls comprise four outputs missing tool names and one incomplete JSON output. The empty array is valid JSON but expresses no decision. Official invalid-output count remains 15 and the matrix's invalid/unknown count remains 16. Original scores, expected behavior, parser rules and result JSON are unchanged.

Diagnostic Change

The behavior analyzer previously classified a matching meta-tool name as `correct_decision_wrong_wording`, even if its required arguments were invalid. It now consults the existing strict parser first. Invalid clarification arguments are labeled `attempted_clarification_invalid_arguments`; the analogous refusal label is `attempted_refusal_invalid_arguments`. Mixed meta/productive calls remain malformed, not accepted boundary decisions.

The quick failure-summary command can print a separate diagnostic table:

```
.venv\Scripts\python scripts\summarize_eval_failures.py `
  results\phase2\qwen-lora\baymax-mix\target-compatible-generator14-act200-anchored-lr1e-4-train1k-val200-epochs2-combined-150.json `
  --output-diagnostics
```

The default failure-reason summary is unchanged. Diagnostics use the current parser only to inspect saved generations. If a recorded rejection/unknown is not reproduced by that parser, it is flagged separately rather than rescored.

Decision and Limits

There is no demonstrated `reason`/`question` defect in the selected targets to fix. Do not rename fields in evaluation outputs or add fixture-specific training examples. Any future output-contract learning experiment should be independently authored, keep class quantities fixed and separate format validity from routing correctness. No new training or prompt change was made during this audit.

Sources

- C result JSON
- C training config and dataset identity
- Shared meta-tool definitions
- Target validation
- Read-only diagnostic classifier
- Failure-summary command

Audit recorded on 12 October 2026. It is a measurement investigation, not a new model run.

Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe

Status: Completed; evaluated on 12 October 2026. Rank 8 remains the reference.

Experiment index

Context

The anchored two-epoch SFT reference remains our reference for balanced ACT / CLARIFY / REFUSE routing. More SFT epochs and the DPO trials have not established a better balanced model. This experiment tests adapter capacity without changing the data or training objective.

What Changes

Setting	A: reference	B: candidate
LoRA rank	8	16
LoRA alpha	16	32
Alpha / rank	2	2
Target modules	q_proj, v_proj	q_proj, v_proj

Rank and alpha change together to retain the existing LoRA scaling ratio. This doubles the trainable adapter parameter count for the same target modules. It does not test adding k_proj/o_proj, lowering the learning rate or adding data.

Fixed Recipe

Setting	Both runs
Initialization	Fresh Qwen/Qwen2.5-1.5B-Instruct, not a continued adapter
Dataset	xlam-baymax-target-compatible-v14-act200-anchored-val200
Dataset revision	329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c
Prompt SHA256	a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468
Raw caps	1,000 training / 200 validation rows
Actual sequence limit	1,024 with the command below

Setting	Both runs
Actual batch / accumulation	1 / 16 with --local-low-vram
Learning rate / epochs	0.0002 / 2
Dropout / seed	0.05 / 1
Scheduler / warmup	Cosine / 0.03

The JSON retains the reference's declared 2,048 context and batch settings; the same command-line overrides reproduce its actual low-VRAM recipe. The reference retained 974 training and 195 validation rows after token filtering. The candidate's final usable-row counts are not embedded in its evaluation JSON; the reference counts must not be presented as newly measured candidate counts.

Run

From the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_rank16.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The candidate writes to its own `models` directory ending in `epochs2_rank16`. Do not overwrite the reference adapter or reduce context to work around memory pressure without recording that as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases, with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The supplied evaluation command uses greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: rank 8	B: rank 16
Full task successes	44/150 (0.293)	46/150 (0.307)
Routing macro-F1	0.514	0.450
Invalid outputs	2/150 (0.013)	7/150 (0.047)
Tool-call accuracy	0.600	0.547
Argument accuracy	0.771	0.785
Clarification wording accuracy	0.075	0.050

Metric	A: rank 8	B: rank 16
Refusal wording accuracy	0.172	0.103
Decision accuracy among known decisions	0.568	0.552
Decision coverage	0.987	0.953
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	1980
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording scores are separate from routing: a correct clarification label can still miss the required question details. Local API cost zero does not mean training, hardware or electricity are free.

Routing Counts

Measurement	A: rank 8	B: rank 16
Correct ACT decisions	57/81	60/81
Correct CLARIFY decisions	11/40	11/40
Correct REFUSE decisions	16/29	8/29
Premature ACT on CLARIFY/REFUSE cases	31/69	42/69

Premature ACT here is the matrix count of productive decisions on boundary requests, not the potentially different count of the scorer's failure-reason label.

A: Rank-8 Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

B: Rank-16 Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	2	5
CLARIFY	23	11	6	0

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
REFUSE	19	0	8	2

Invalid/unknown is the decision-matrix category. Invalid-output rate is computed separately from parser failures; the counts happen to match in these two runs.

Interpretation and Decision

The candidate completed two more tasks and slightly improved argument accuracy, but routing macro-F1 fell by 0.064. Correct refusals halved, premature ACT rose by eleven cases, and invalid outputs increased from two to seven. Correct clarification routing did not improve. Rank 16 is not a better balanced replacement.

Keep A, the rank-8 two-epoch SFT model, as the reference. This run does not support insufficient adapter capacity as the main problem. It does not prove capacity never matters, and a single-seed result cannot establish the cause of the regression or whether a two-task difference is reproducible.

The next test recommended after B was C: return to rank 8 / alpha 16 and lower only the learning rate from 2e-4 to 1e-4. That test has now been run; see Experiment C for its results. B remains a separate completed experiment, not part of a combined rank-and-learning-rate change.

Sources and Reproducibility

Artifact	Purpose
A evaluation JSON	Original reference results
B evaluation JSON	Rank-16 results, recorded at 2026-10-12T16:58:08.530056+00:00
B training config	Candidate settings and data identity
Comparison helper	Reads saved metrics and derives routing from recorded decisions

B canonical config SHA256: `a09ca9973edbc9127f7641cc5e10cb809baf18f13a3504c011350835fc0bd663`.

Numbers were checked against the saved result files using the comparison helper. No training, inference or result-file rewriting was performed to record this note. These repeatedly inspected suites are development comparisons, not an untouched final test. Exact base-model revision and complete generation/training metadata are not embedded in the result JSON, which limits retrospective verification.

Experiment C: Lower Learning Rate at Two Epochs

Status: Completed; evaluated on 12 October 2026. A remains the balanced reference.

Experiment index

Question and Context

Does halving the learning rate improve balanced ACT / CLARIFY / REFUSE routing without sacrificing productive task completion?

Experiment B increased adapter rank but worsened routing and validity. C starts from A's original recipe, not B's rank-16 settings and not a saved trained adapter. Smaller updates preserving decision boundaries is a hypothesis to test, not an established explanation of earlier runs.

Controlled Change

Setting	A: reference	C: candidate
Peak learning rate	0.0002	0.0001
LoRA rank / alpha	8 / 16	8 / 16
Target modules	q_proj, v_proj	q_proj, v_proj
SFT epochs	2	2
Initialization	Fresh Qwen base + fresh LoRA	Fresh Qwen base + fresh LoRA

Only the learning rate changes, apart from the run name, config hash and output paths needed to keep the experiment separate. Cosine scheduling and 3% warmup remain unchanged; C halves the schedule's learning-rate amplitude.

Fixed Data and Actual Overrides

- Base model: Qwen/Qwen2.5-1.5B-Instruct.
- Data: xlam-baymax-target-compatible-v14-act200-anchored-val200, unchanged splits.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual sequence limit: 1,024; train/eval batch one; gradient accumulation 16.
- Seed 1, dropout 0.05, assistant-only token cross-entropy and the same masking.

The command below applies the same low-VRAM overrides as A; the JSON retains A's declared context and batch settings. A retained 974 train and 195 validation rows after token filtering. C's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of C.

Training Command

Run from the repository root in PowerShell:

```
$config = "configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_lr1e-4.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new output directory ends in `epochs2_lr1e-4`; neither A nor B is overwritten.

Results

Both runs use the established 150-case v1 + v2 development suite: 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: learning rate 0.0002	C: learning rate 0.0001
Full task successes	44/150 (0.293)	45/150 (0.300)
Routing macro-F1	0.514	0.436
Invalid outputs	2/150 (0.013)	15/150 (0.100)
Tool-call accuracy	0.600	0.513
Argument accuracy	0.771	0.726
Clarification wording accuracy	0.075	0.050
Refusal wording accuracy	0.172	0.172
Decision accuracy among known decisions	0.568	0.560
Decision coverage	0.987	0.893
Unavailable-tool hallucination rate	0.020	0.010
Average latency (ms)	1828	2492
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording metrics are not routing metrics. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	C
Correct ACT decisions	57/81	58/81
Correct CLARIFY decisions	11/40	4/40
Correct REFUSE decisions	16/29	13/29
Premature ACT on boundary cases	31/69	37/69
Invalid/unknown decisions	2/150	16/150

Premature ACT is derived from the matrix's CLARIFY/REFUSE rows, not necessarily the count of the scorer's identically themed failure-reason label.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

C: Lower-Learning-Rate Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	58	13	5	5
CLARIFY	24	4	4	8
REFUSE	13	0	13	3

Invalid/Unknown Output Inspection

Observed issue	Responses
request_clarification used reason instead of question	10
Calls omitted tool names	4
Incomplete JSON	1
Empty array with no decision	1

The ten wrong-field calls include six expected CLARIFY cases, two ACT cases and two REFUSE cases. An attempted clarification is not necessarily correct routing. The empty array parses but expresses no decision; that explains 15 invalid outputs versus 16 invalid/unknown decisions. None were repaired or rescored.

The target-contract audit checked the selected 1,000 raw training and 200 validation targets. All 100 training and 30 validation clarification targets correctly use question. This does not establish that the model learned the contract or that every target is semantically perfect.

Interpretation and Decision

C gained one full task success, but macro-F1, argument accuracy and output validity worsened. The lower rate did not preserve the boundaries better in this run. Keep A as the balanced reference; do not promote C on its one-task gain.

Some apparent clarification losses are invalid expression of an attempted clarification, not only selection of the wrong decision. Output-contract learning should be investigated separately without weakening the strict parser. This experiment alone does not prove that the lower rate caused undertraining or that more epochs would resolve the defects.

Provenance and Limits

- C evaluation JSON: recorded 2026-10-12T17:34:32.282617+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- C canonical config SHA256: 80b81f44f0daf9f4d18ba8f29eb7139a216d72c5ecec6b5839ffd77e40eeee02.

Metrics and matrices were checked using the existing comparison helper on saved artifacts. No training, inference or result modification was performed to write this note. Results are single-seed observations on repeatedly inspected development suites, not an untouched final benchmark. Exact base-model revision and complete training/generation metadata are not embedded in the result JSON.

Supporting Files

- Candidate config
- A: selected rank-8 two-epoch reference

Experiment E: QKVO LoRA on All Four Attention Projections

Status: Completed; evaluated on 12 October 2026. Task-completion candidate, not the balanced reference.

Experiment index

Question and Controlled Change

Does adapting key and output projections alongside query and value projections improve task completion and balanced routing under A's original two-epoch recipe?

Setting	A: reference	E: candidate
Target modules	q_proj, v_proj	q_proj, k_proj, v_proj, o_proj
Rank / alpha	8 / 16	8 / 16
Learning rate	0.0002	0.0002
Epochs	2	2
Initialization	Fresh pretrained Qwen + fresh LoRA	Fresh pretrained Qwen + fresh LoRA

Only the target-module list changes, apart from run identity and output paths. E is not the rank-16 recipe or the lower-learning-rate recipe. Broader coverage also increases trainable adapter parameters: this is not a constant-parameter coverage comparison. Its result cannot attribute a gain solely to module placement.

Fixed Recipe

- Base model: Qwen/Qwen2.5-1.5B-Instruct; pretrained weights are not reinitialized.
- Frozen data: xlam-baymax-target-compatible-v14-act200-anchored-val200.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual command overrides: context 1,024, train/eval batch one, accumulation 16.
- Seed 1, dropout 0.05, cosine scheduling with 3% warmup, existing PEFT initialization.
- The same assistant-label masking, target-contract checks and strict inference parser.

The JSON retains A's declared context and batch settings. The command applies the same actual low-VRAM overrides as A. A retained 974 training and 195 validation rows. E's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of E.

Training Command

Run from the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The separate output directory ends in `epochs2_qkvo`. No previous model is overwritten. Additional adapted modules may increase memory pressure. Do not silently reduce sequence length or change quantization to finish this run; record any such change as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, 256 new tokens and the existing strict parser.

Metric	A: query/value only	E: query/key/value/output
Full task successes	44/150 (0.293)	52/150 (0.347)
Routing macro-F1	0.514	0.437
Invalid outputs	2/150 (0.013)	14/150 (0.093)
Tool-call accuracy	0.600	0.527
Argument accuracy	0.771	0.748
Clarification wording accuracy	0.075	0.113
Refusal wording accuracy	0.172	0.086
Decision accuracy among known decisions	0.568	0.581
Decision coverage	0.987	0.907
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	3794
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording checks are separate from routing. Decision accuracy excludes unknown decisions; read it alongside coverage. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	E
Correct ACT decisions	57/81	63/81
Correct CLARIFY decisions	11/40	10/40
Correct REFUSE decisions	16/29	6/29
Premature ACT on boundary cases	31/69	40/69
Invalid/unknown decisions	2/150	14/150

Premature ACT here is the count derived from the CLARIFY/REFUSE matrix rows, not necessarily the scorer's failure-reason count.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures routing correctness, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

E: QKVO Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	63	12	2	4
CLARIFY	27	10	2	1
REFUSE	13	1	6	9

Invalid-Output Inspection

Observed issue	Expected cases	Responses
refuse_request put reason outside arguments	REFUSE	6
request_clarification had empty arguments	CLARIFY 1, REFUSE 2	3
Productive calls omitted or separated names from arguments	ACT	4
Other malformed output	REFUSE	1

Six of the nine invalid REFUSE-case outputs selected the refusal meta-tool name, but supplied reason at the call's top level instead of inside arguments:

```
{"name": "refuse_request", "arguments": {}, "reason": "Unsupported operation."}
```

The contract requires:

```
{"name": "refuse_request", "arguments": {"reason": "Unsupported operation."}}
```

These are generic illustrations of the nesting defect, not repaired benchmark responses. Selecting refuse_request does not establish that its explanation meets the scenario's wording requirements. No rejected response received success credit.

Interpretation and Decision

E completed eight more tasks than A, the largest task-success gain among the A/B/C/E ablations. However, macro-F1 fell, invalid outputs rose, and latency was about 2.1 times A's measured latency. E is not a better balanced replacement.

Keep A as the balanced reference and E as a task-completion candidate. A substantial part of E's observed refusal loss involves invalid expression of an attempted refusal. This supports investigating output-contract learning separately; it does not erase the other incorrect decisions or establish a hypothetical repaired score.

The next proposed test is an independently authored output-contract learning experiment with the policy class quantities and other training settings held fixed. No data adjustment or new run has been made by this documentation task.

Provenance and Limits

- E evaluation JSON: recorded 2026-10-12T19:19:35.198791+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- E canonical config SHA256: ee2c1b9efdec057796d0c0a9d16b20f75763151fcc34e6fe95ec004bc3bf00e4.

Metrics and matrices were checked with the existing comparison helper. Raw generations were inspected read-only. Original scores, results and parser rules remain unchanged. These are single-seed development observations, not an untouched final test. Adding modules also increases adapter size, so this does not isolate placement at equal parameter count. Full training/generation metadata and exact base-model revision are not embedded in the result JSON.

Supporting Files

- E config
- A: selected reference

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.